AQUA-SENSE

AI-POWERED IOT SYSTEM FOR PUBLIC WATER QUALITY MONITORING & PREDICTIVE ALERTS

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INTRODUCTION

- Our project is AquaSense, an IoT and AI-based water quality monitoring system.
- It uses sensors to measure parameters like pH, turbidity, temperature and TDS.
- The data is sent in real time to the cloud using LoRa communication.
- An AI model (CNN-LSTM) analyzes the data to detect contamination and predict water quality.
- The results, alerts, and recommendations are shown in a mobile application.
- The problem is that traditional water testing is slow and not real-time.
- This delay can affect public health and there is no transparency or instant alerts.
- So, our objective is to develop a real-time monitoring system.
- We use AI for early detection, a mobile app for alerts, and improve communication between authorities and users. Overall, the system provides a proactive solution for water safety.

ABSTRACT

Aquasense is an AI-based, IoT water quality monitoring system developed to improve water safety management. It uses sensors to measure important parameters such as pH, turbidity, and temperature. The collected data is sent in real time to the cloud using LoRa communication. A CNN-LSTM model analyzes this data to check water quality and predict possible contamination. The results and simple recommendations are shown to authorities and users through a mobile application. By replacing slow laboratory testing with real-time monitoring, Aquasense helps in early problem detection, better decision-making, and improved public trust, ensuring safe and healthy water for communities.

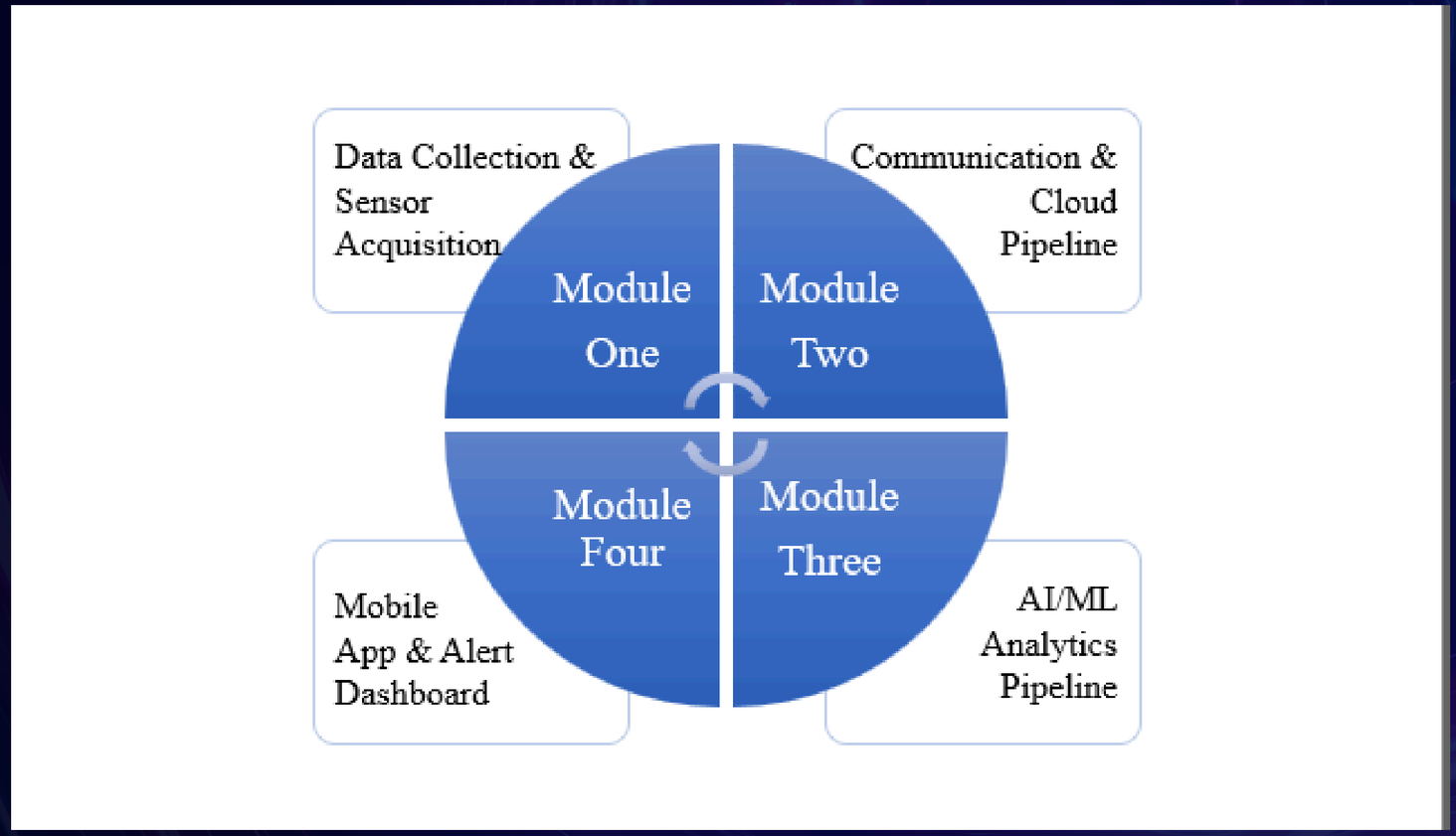
PROBLEM STATEMENT

- Traditional water monitoring is slow, manual, and not real-time.
- Delays in detection can risk public health.
- Late detection leads to ineffective response actions.
- Lack of transparency, data access, and instant alerts.
- Need for a real-time, automated, and proactive monitoring system.

OBJECTIVES

- To design a real-time IoT sensor network for continuous water quality monitoring.
- To implement a predictive AI model for early contamination detection.
- To develop a user-friendly mobile app for data visualization and alerts.
- To establish a transparent communication system between authorities and citizens.
- To create a proactive, cost-effective solution for water safety management.

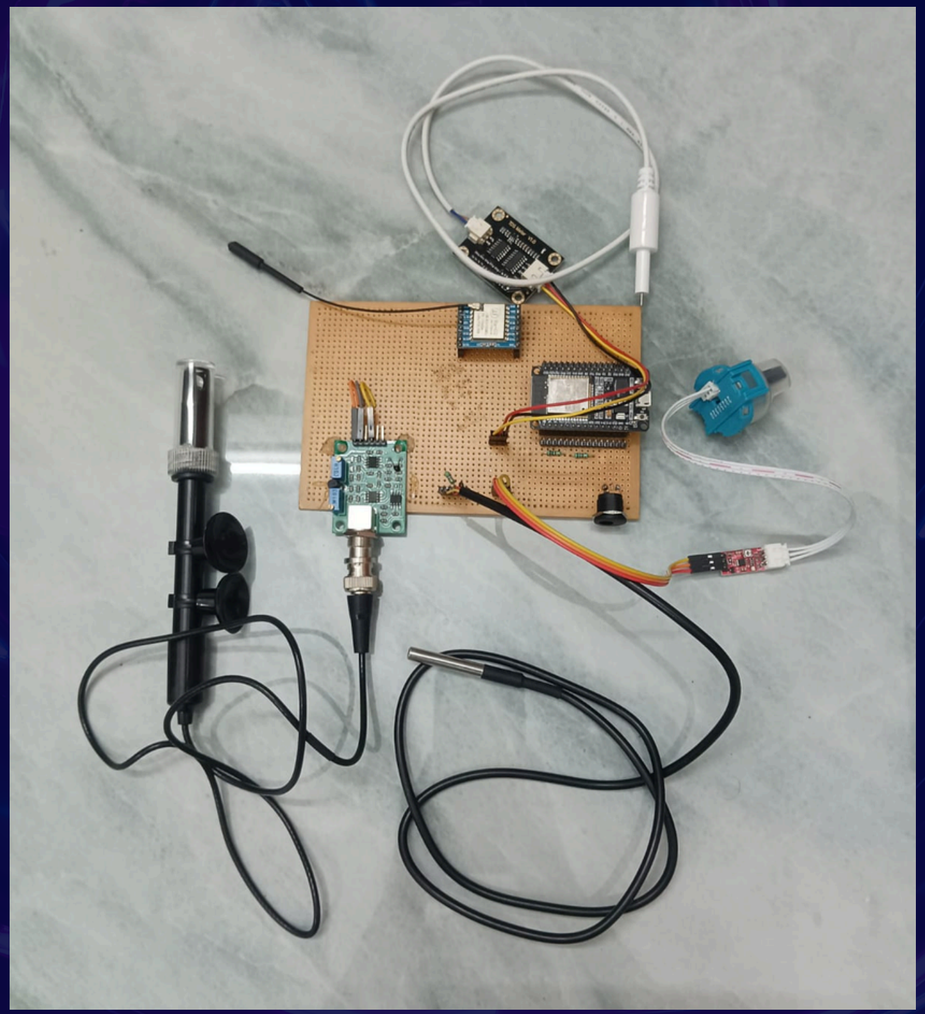
MAIN MODULES OF THE PROJECT



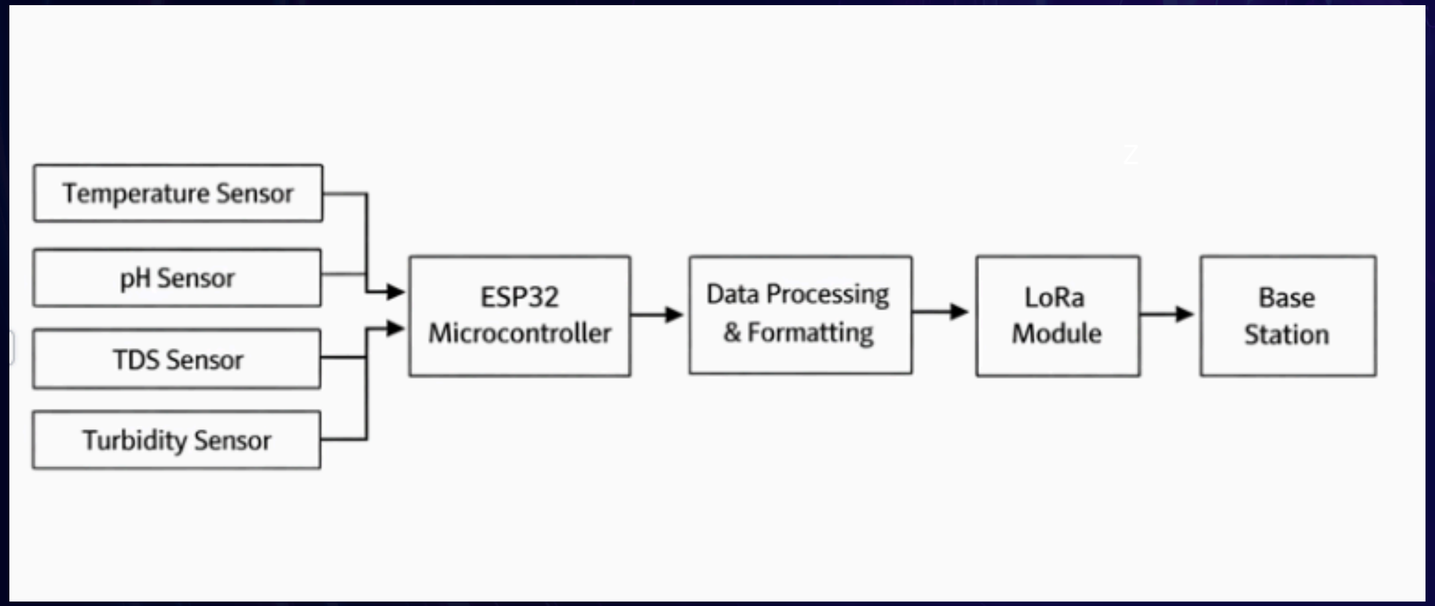
ARCHITECTURE 1: DATA COLLECTION & SENSOR ACQUISITION

SENDER-SIDE HARDWARE PROTOTYPE

- ESP32 Development Board
- Temperature Sensor (DS18B20)
- pH Sensor
- TDS Sensor
- Turbidity Sensor
- LoRa Communication Module (SX1278)
- Power Supply Unit
- Connecting Wires and Breadboard



SENSOR DATA ACQUISITION WORKFLOW

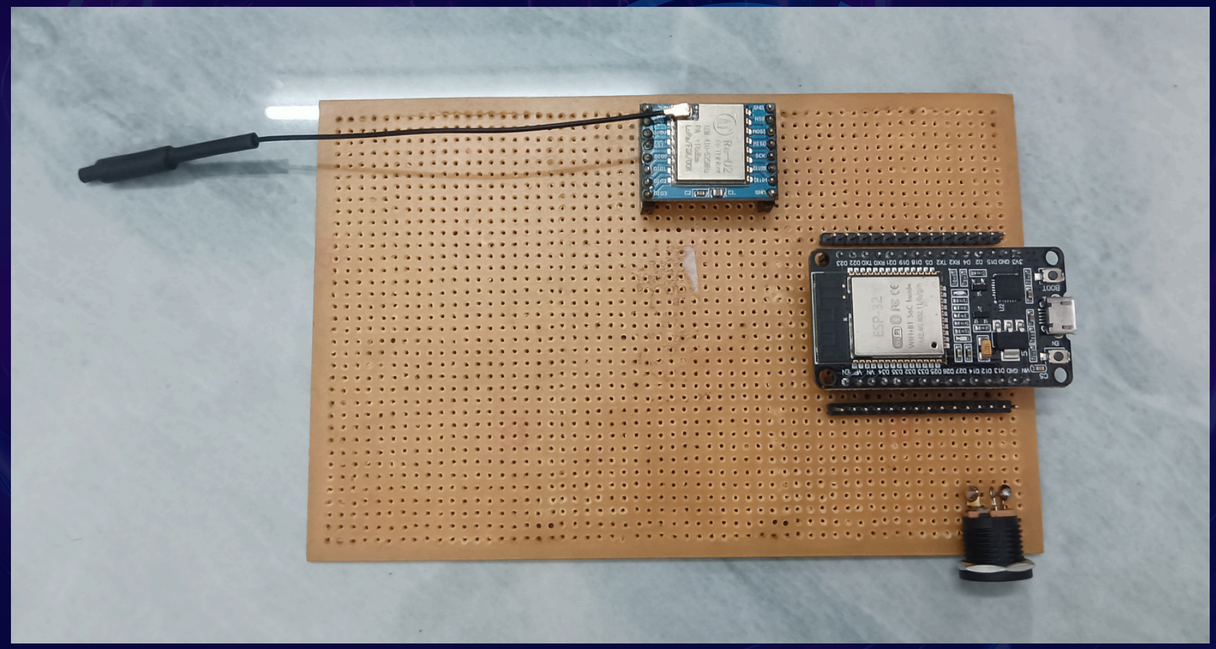


- Water quality sensors continuously monitor key parameters such as pH, Temperature, TDS, and Turbidity
- Raw analog outputs from sensors are interfaced with the ESP32 microcontroller
- ESP32 converts analog signals into digital values for processing
- Initial calibration and filtering are applied to improve reading accuracy
- Sensor values are formatted into structured data packets
- Processed data is forwarded to the LoRa module for wireless transmission

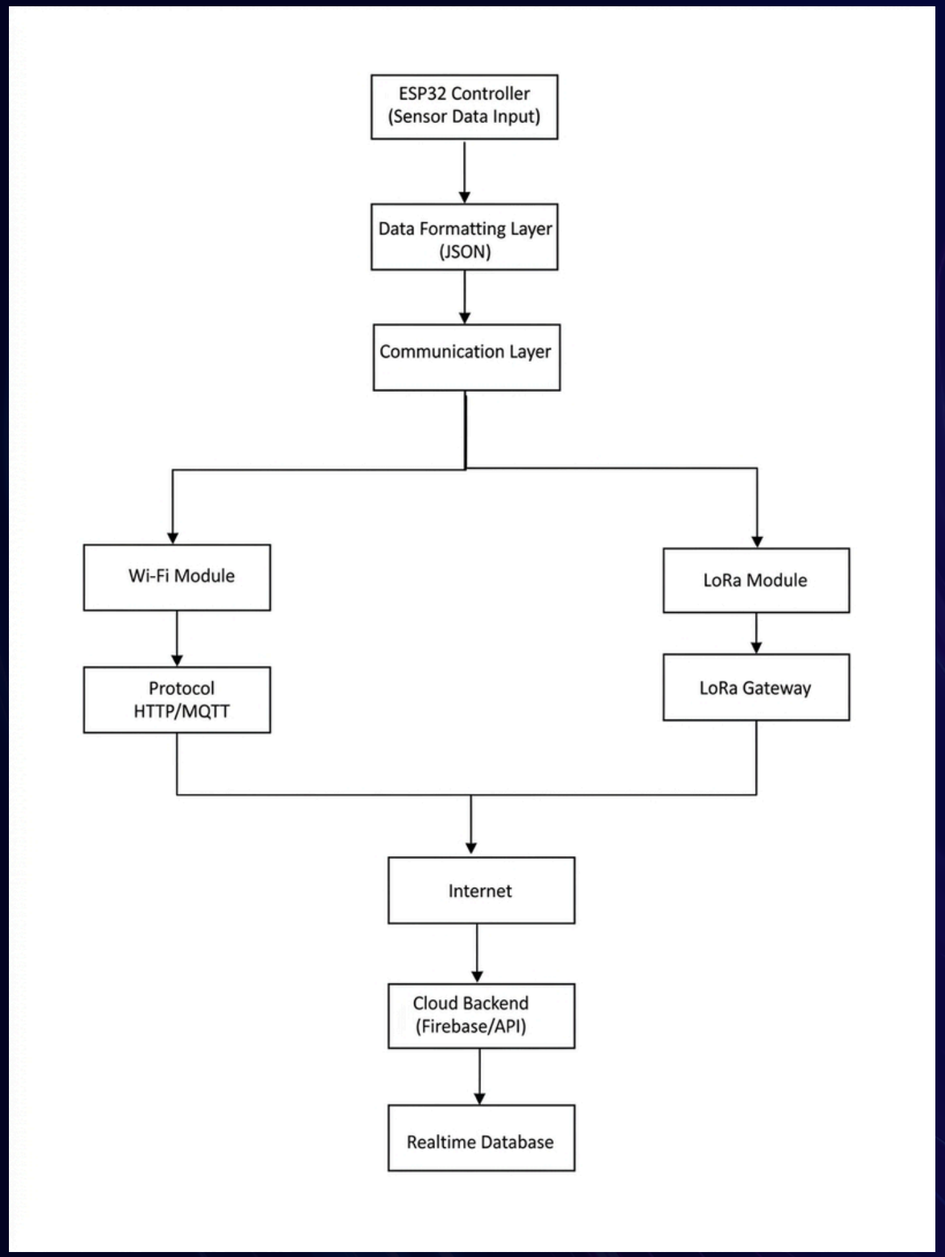
ARCHITECTURE 2: COMMUNICATION & CLOUD PIPELINE

HARDWARE COMPONENTS USED

- ESP32 Development Board
- LoRa Communication Module (SX1278)
- Power Supply Unit



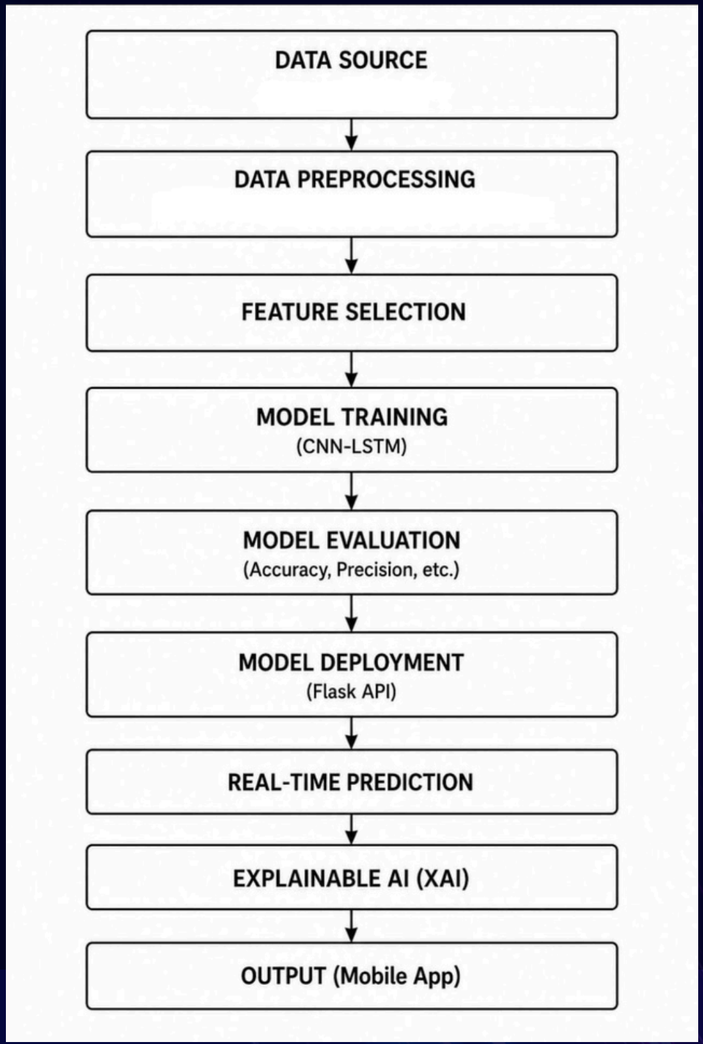
RECEIVER-SIDE HARDWARE PROTOTYPE



WORKFLOW

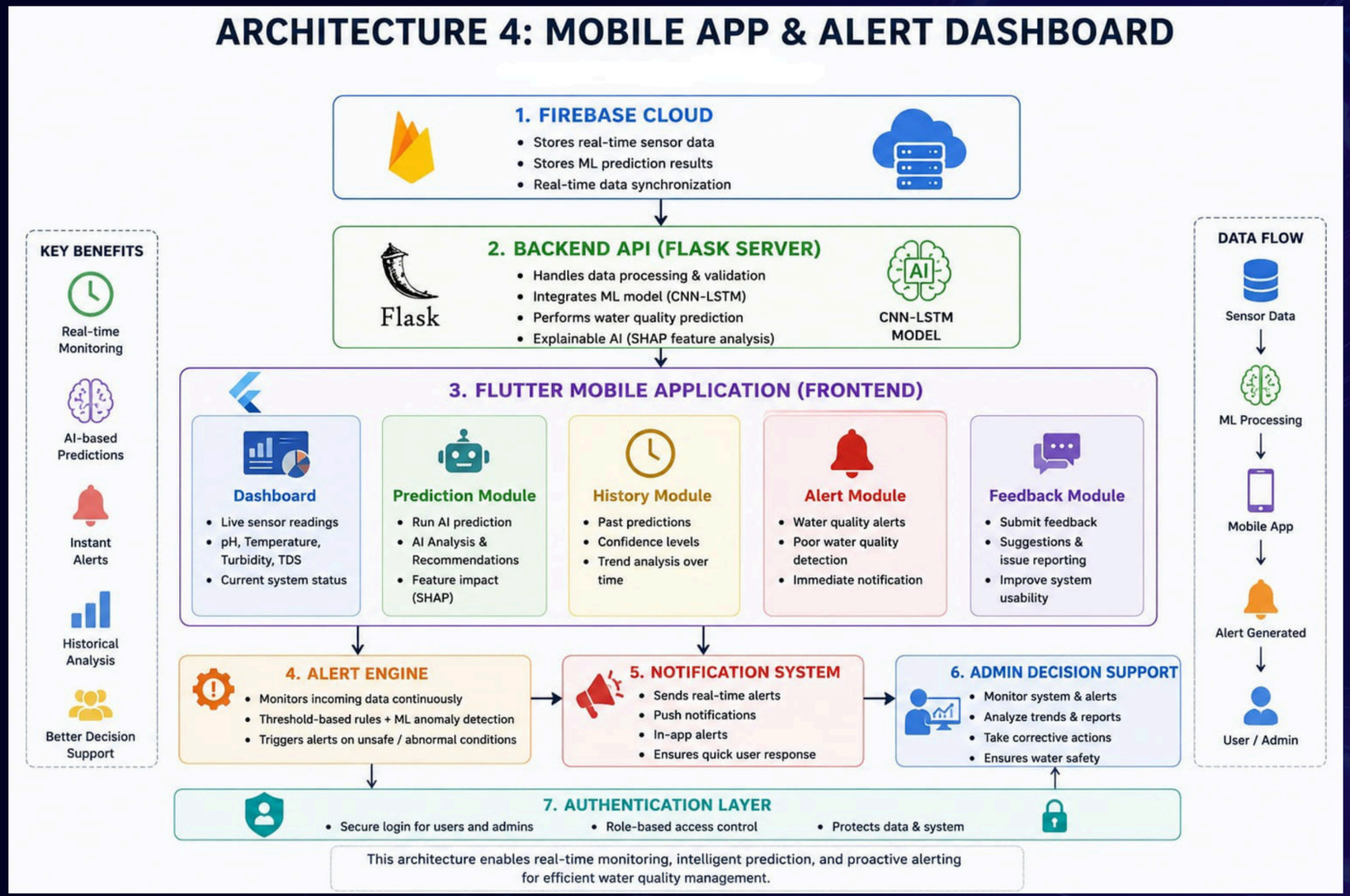
- ESP32 receives sensor data and prepares it for transmission
- Data is formatted into JSON for structured communication
- Communication layer selects between WiFi and LoRa modules
- WiFi module transmits data using HTTP or MQTT protocols
- HTTP supports request–response communication with cloud APIs
- MQTT enables lightweight, real-time data transmission
- LoRa module is used for long-range, low-power communication
- LoRa data is forwarded to a gateway before reaching the internet
- Gateway converts LoRa signals into IP-based data
- Internet transfers data to cloud backend (Firebase / API)
- Cloud backend processes and stores data in realtime database

ARCHITECTURE 3: AI/ML ANALYTICS PIPELINE



- The AI/ML pipeline starts with collecting a dataset containing water quality parameters such as pH, turbidity, temperature and TDS.
- The data is preprocessed by cleaning, normalizing, and encoding labels.
- Important features are selected to improve model accuracy.
- A CNN-LSTM model is trained for prediction.
- The model is evaluated using accuracy, precision, recall, and F1-score.
- The trained model is deployed using a Flask API.
- Real-time prediction is performed using live sensor data.
- Explainable AI provides reasons and recommendations for predictions.
- The output is displayed in the mobile app as safe/unsafe water with alerts.

Architecture 4 – Mobile App & Alert Dashboard



Firebase Cloud

- Stores sensor & prediction data
- Real-time data sync

Backend API (Flask)

- Data processing & validation
- ML prediction (CNN-LSTM)
- SHAP for feature impact

Flutter Mobile App

- Real-time data display
- User-friendly interface

Alert Engine

- Triggers alerts
- Threshold-based rules

Notification System

- Push notifications / in-app alerts
- Immediate user awareness

Admin Decision Support

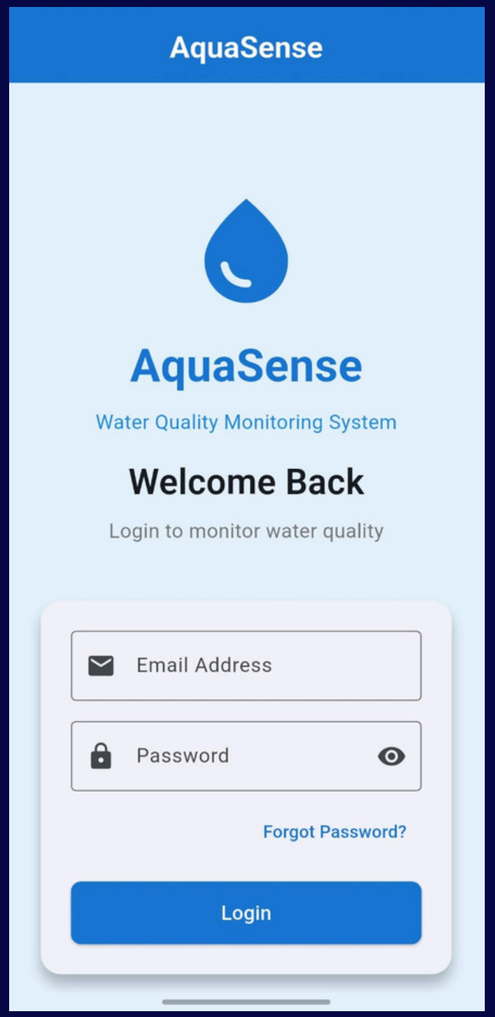
- Monitor system & alerts

Authentication Layer

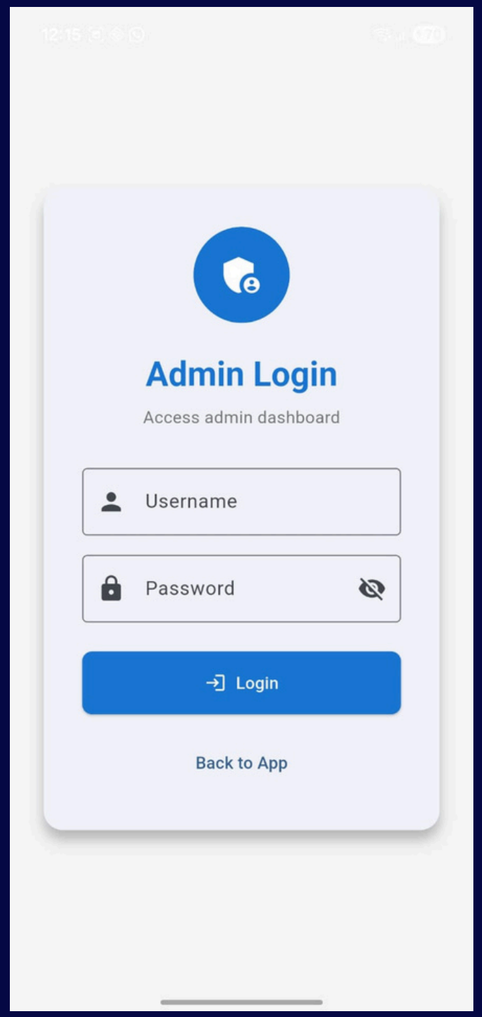
- Role-based access control
- Secure login

APPLICATION INTERFACE PREVIEW

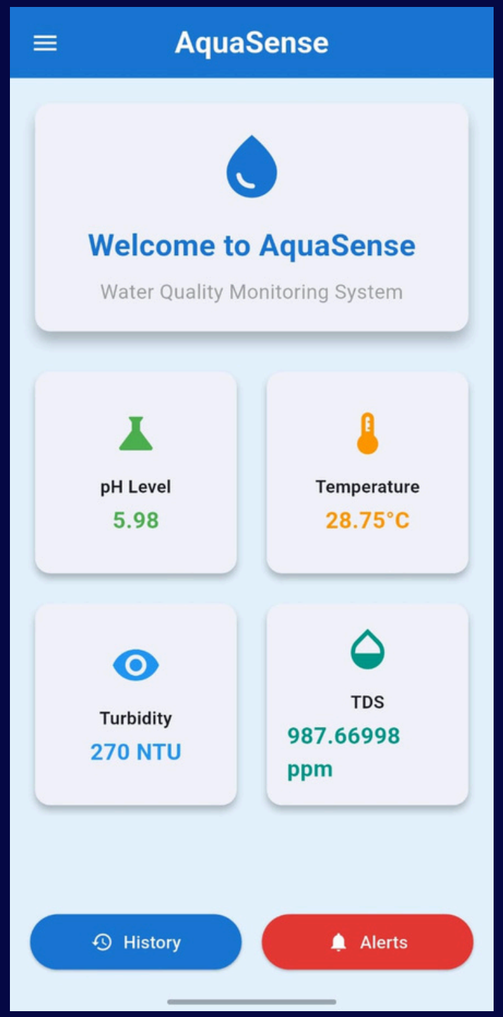
Login Page



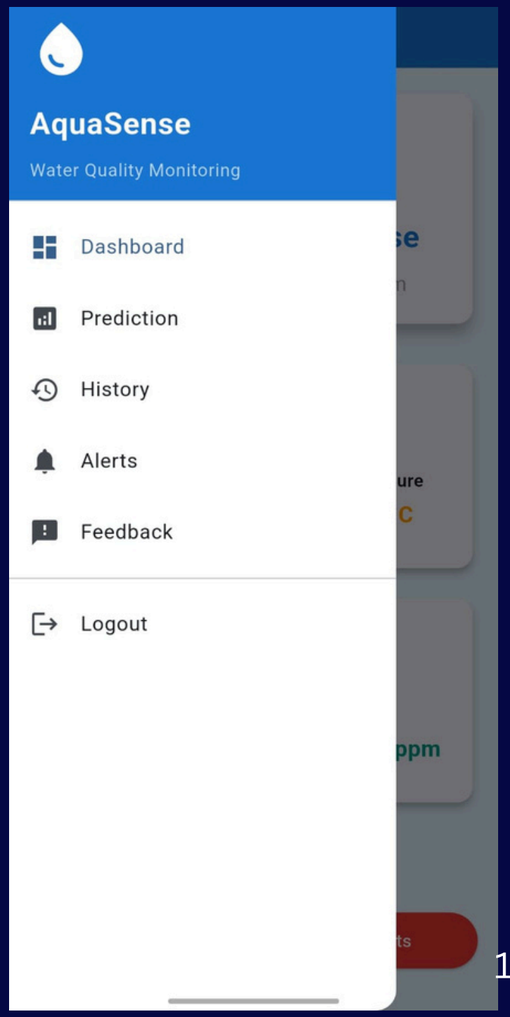
Admin Login



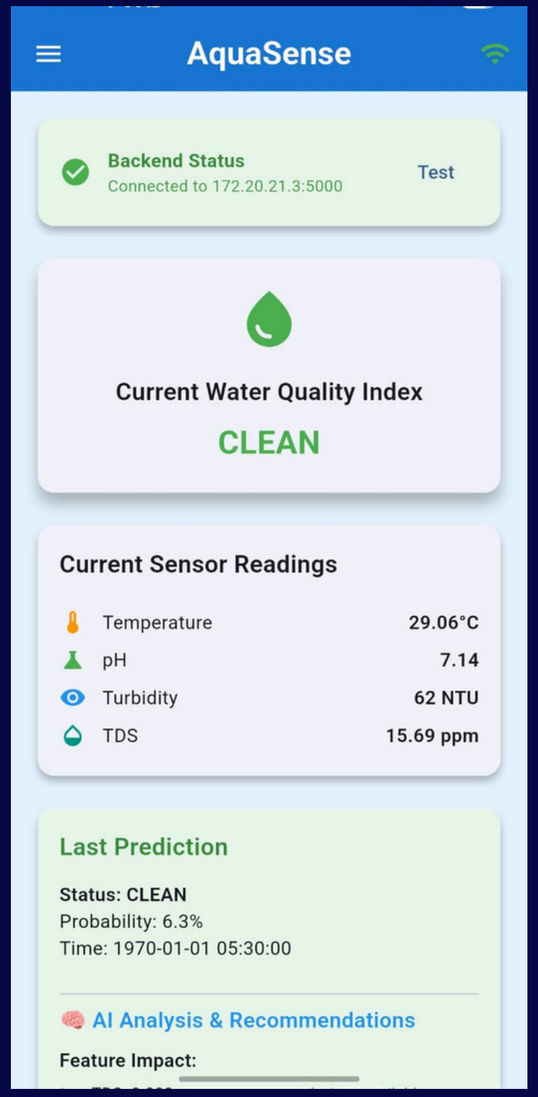
Home Page

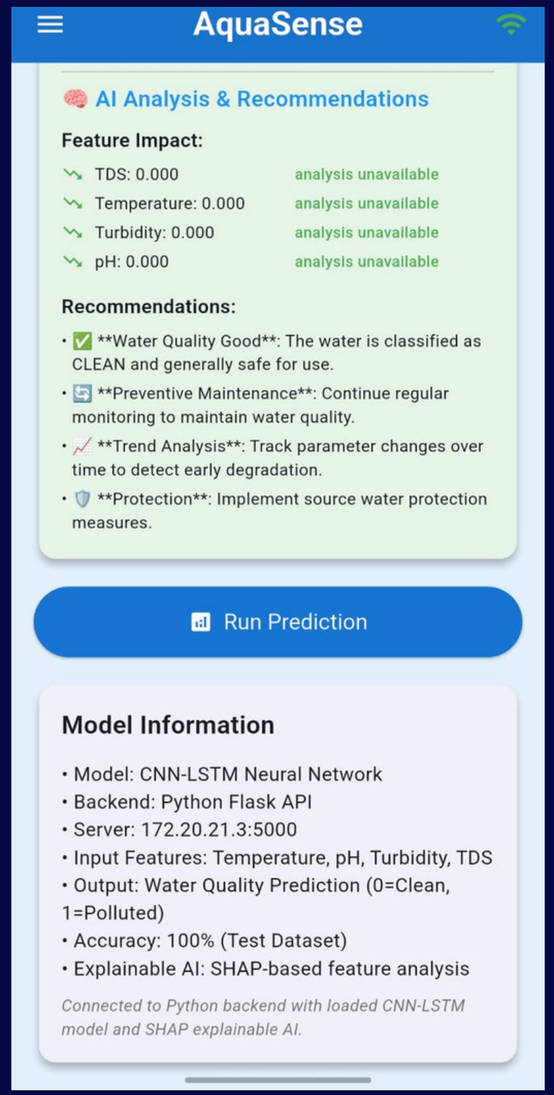


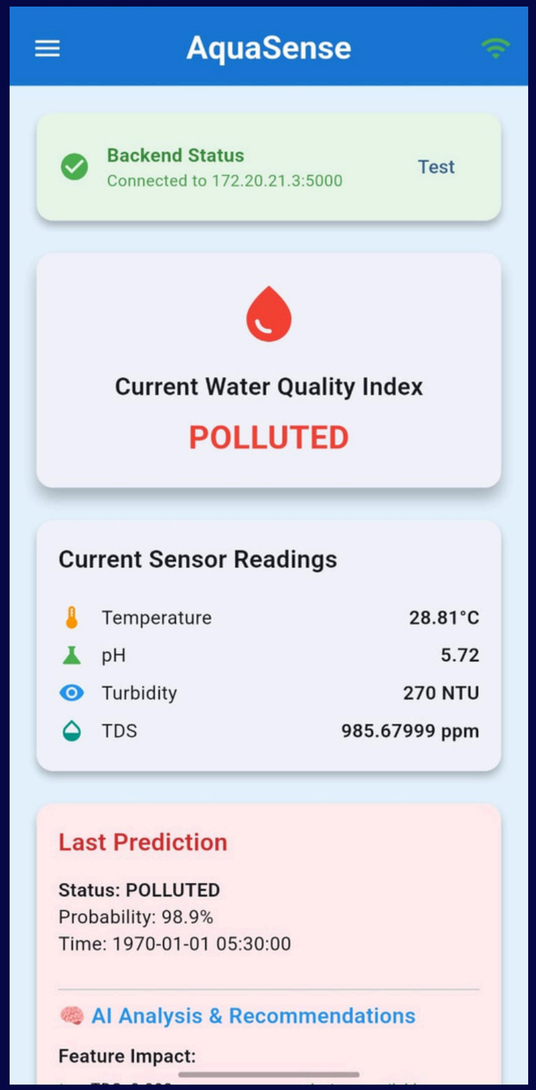
Navigation Bar

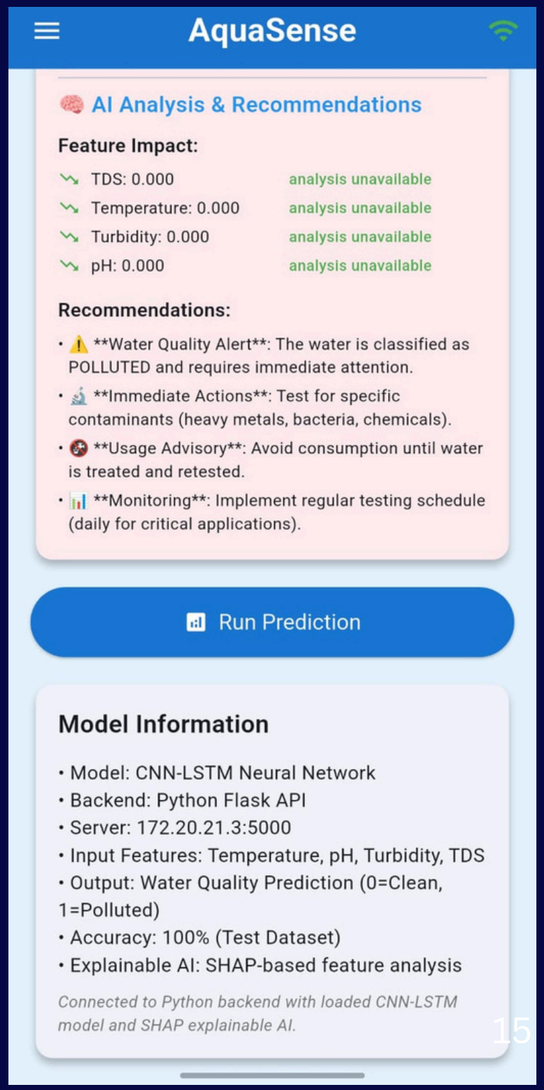


Prediction Page

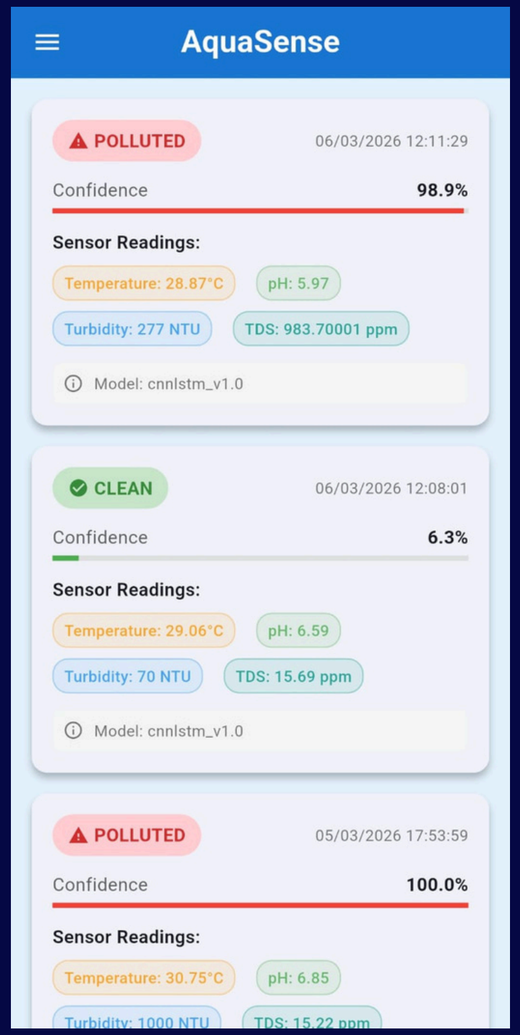




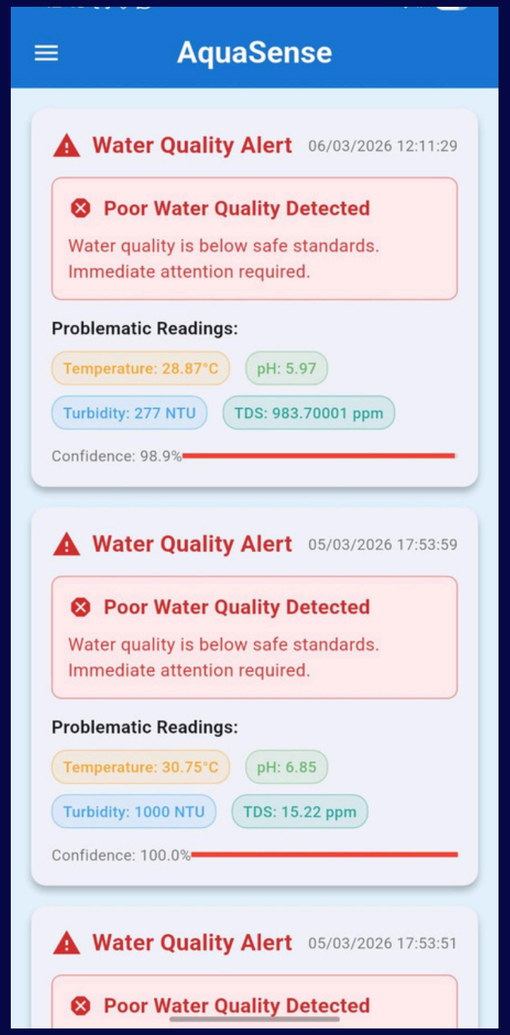




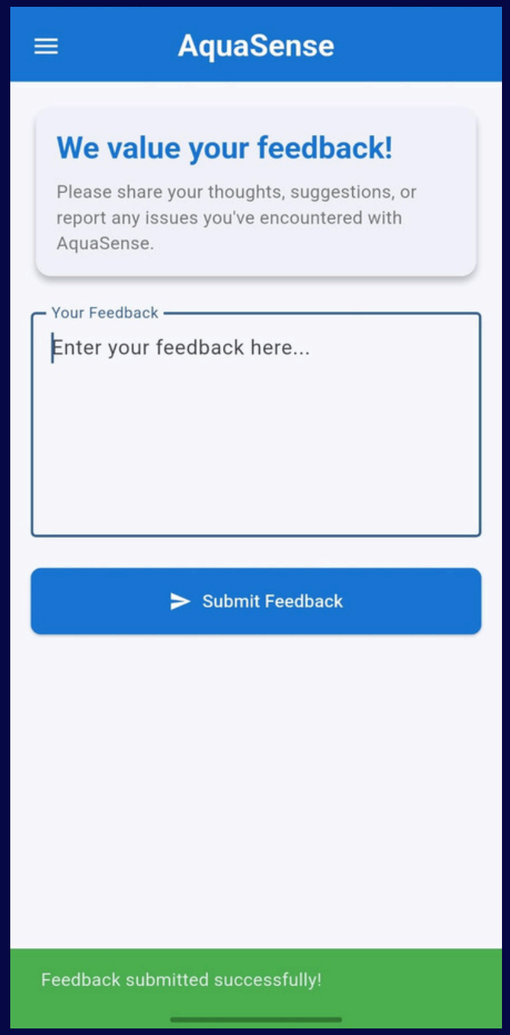
History Page



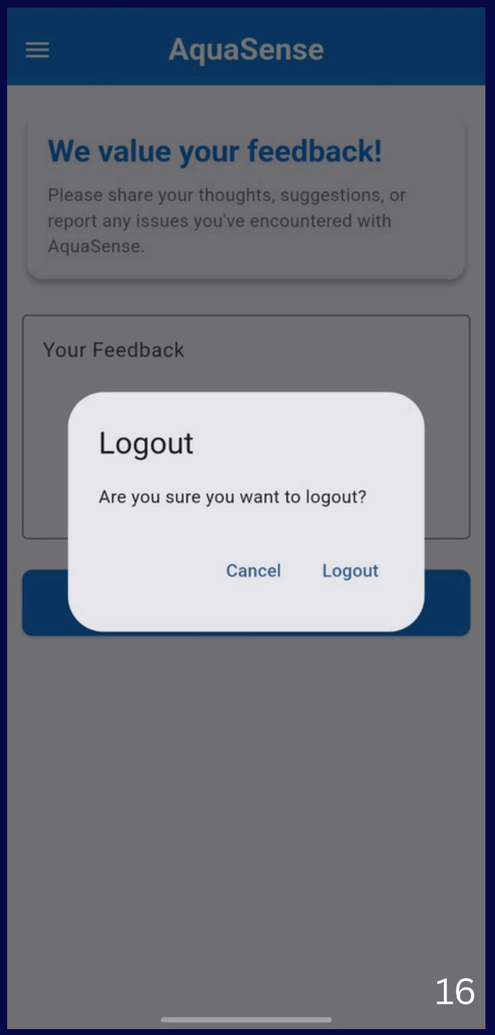
Alert Page



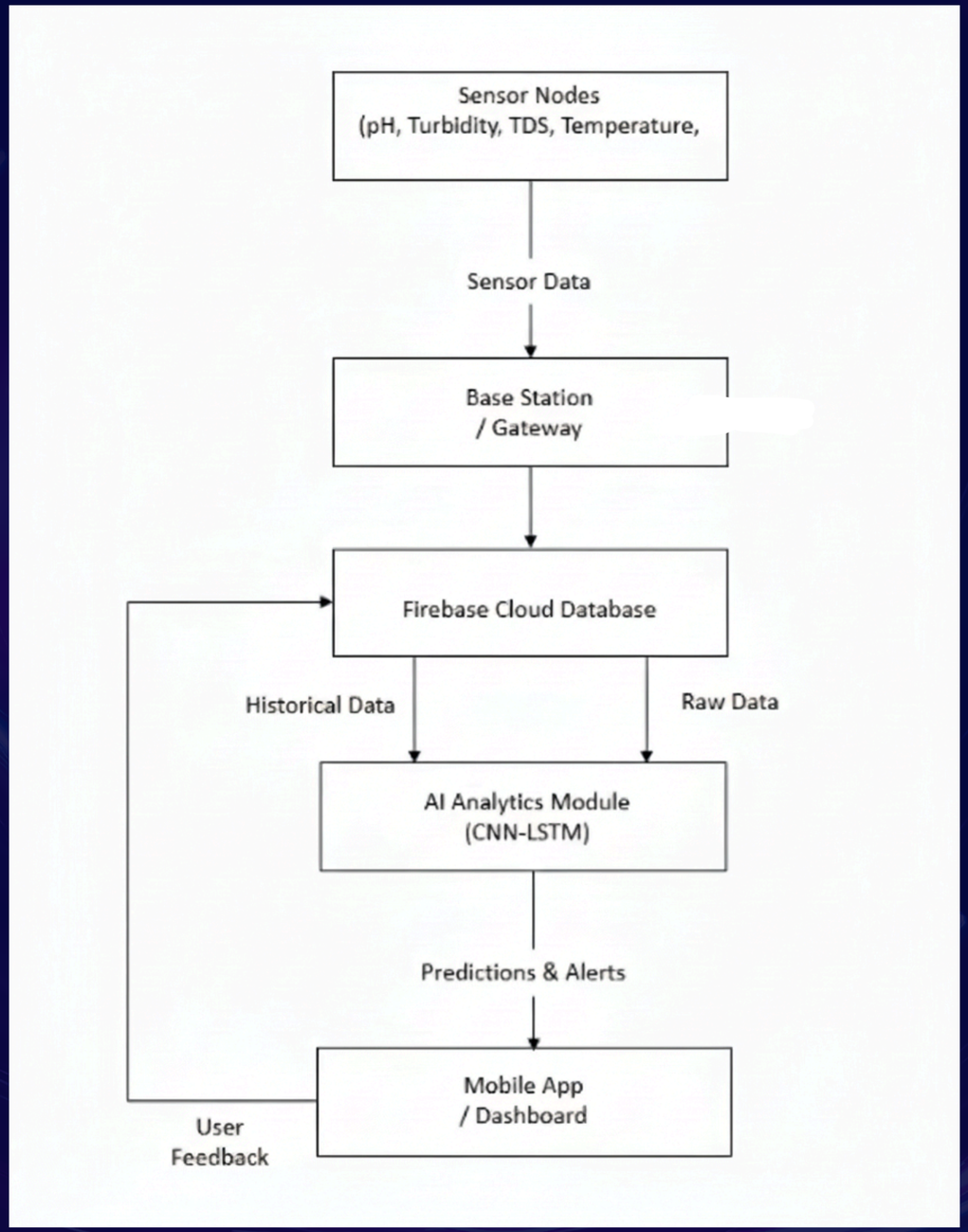
Feedback Page



Logout Page



SYSTEM ARCHITECTURE



NOVELTY OF AQUA-SENSE

AI-Powered Predictive Monitoring

- Unlike traditional systems that only display current water conditions, Aqua-Sense uses a CNN-LSTM hybrid AI model to predict future contamination risks.

Real-Time Cloud-Based Monitoring

- Water quality data is continuously synchronized to Firebase Cloud, enabling instant access, storage, and visualization through mobile applications.

Public Transparency and Accessibility

- Aqua-Sense provides real-time water quality information and alert notifications to both authorities and public users through a Flutter-based mobile app.

Explainable AI (XAI) Support

- The system not only predicts contamination but also provides interpretable insights into the factors influencing water quality predictions.

Integration of IoT and LoRa Technology

- The system combines IoT sensors with LoRa communication to achieve real-time monitoring with long-range and low-power data transmission.

LIMITATIONS

Limited Parameter Monitoring

- The current prototype monitors only selected parameters such as pH, turbidity, temperature, and TDS, while advanced biological contaminants are not detected.

Hardware Durability Issues

- Sensors and electronic components deployed in outdoor environments may be affected by water exposure, corrosion, and harsh environmental conditions.

Power Dependency

- Continuous system operation depends on a stable power supply, which can be challenging in remote or rural deployment areas.

Sensor Maintenance Requirement

- The sensors require periodic cleaning and recalibration to maintain reliable and accurate measurements over long-term usage.

Future Scope

Advanced AI & Predictive Intelligence

- Integrate deep learning models for higher accuracy across diverse water conditions
- Enable self-learning models that improve continuously with incoming sensor data

IoT & Hardware Expansion

- Support multi-sensor integration (heavy metals, bacteria detection, turbidity, etc.)
- Improve battery efficiency & solar-powered modules for sustainability

Cloud & Communication Optimization

- Transition to a fully scalable cloud architecture (handling large-scale deployments)
- Implement data redundancy & failover systems for reliability

Enhance User Experience

- Introduce offline mode with sync capabilities
- Provide multi-language support for wider accessibility

Security & Data Governance

- Implement end-to-end encryption for IoT communication
- Add user authentication & role-based access control

Research & Innovation Extensions

- Incorporate satellite/weather data correlation for better predictions
- Expand into climate impact analysis on water quality

CONCLUSION

Problem Addressed

- Lack of real-time, accessible water quality monitoring
- Difficulty in identifying unsafe water conditions early

Our Solution

- Designed an IoT + AI-based smart monitoring system
- Integrated sensor data, cloud pipeline, and ML prediction
- Delivered insights through a user-friendly mobile interface

Key Achievements

- Built a working end-to-end architecture (Sensor → Cloud → AI → App)
- Enabled accurate water quality prediction
- Implemented Explainable AI for transparency
- Established a scalable system foundation

Impact & Value

- Supports public health and safety
- Useful for smart cities, rural areas, and industries
- Promotes data-driven environmental monitoring

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THANK YOU!